Optimized Power Method
(Preview: Only Page 1)

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Introduction

The purpose of this report is to compare the results obtained using Basic Linear Algebra Subprograms (BLAS) in the utility functions with those from HW 3b, where we implemented a parallel version of the power method in C. For context, the power method is used to compute approximations to the largest eigenvalue in magnitude and associated eigenvector of a matrix. For this assignment, we are using the utility functions from the previous report that included a parallel version of the power method.

We used the standard initial guess for the eigenvector $x \in R^n$ with components $x_j = 1/\sqrt{n}$ for all $0 \leq j < n$. Our power method function returns the eigenvalue approximation λ, the eigenvector approximation x, and the number of iterations taken. We used the same n-values that we used in HW 3b (1024, 2048, 4096, 8192, 16384, 32768, 65536, and 131072). After conducting a test for each n-value, we compared the new λ values to the values we obtained in HW 3b and verified that they matched.

Methodology

Implementing BLAS

We used a strategy involving #ifdef BLAS ... #else ... #endif to implement an alternative version using BLAS in our existing utility functions. We revisited all of our previous utility functions and made the following changes:

- Added an MPI_COMM_WORLD parameter to the normE() function
- Added a BLAS implementation to the serial_dot() function
- Edited the Makefile to handle BLAS
- Added the BLAS2 implementation to the matvec() function in utilities.c (which is the highest order BLAS we can use for matrix-vector multiplication)

Immediately, we observed that adding the MPI_COMM_WORLD parameter to the normE function was beneficial for parallel processing. We adjusted the corresponding function calls for this as well to include the new parameter. Here is how the updated function appeared: